HOMEWORK #2

F-16 Trim and Linearization

UUM 523E – Advanced Flight Dynamics · Fall 2026-2027

Homework at a glance	
Due date	Nov 13, 2026
Points	Problem 1: 50 points; Problem 2: 50 points (part a: 25, part b: 25)
Report	At most 8 pages, submitted through Ninova
MATLAB code	Trim and linearization in m-files (may call the HW1 Simulink models); trim with `fmincon`
Simulink models	If used, saved in MATLAB R2023a
Results file	`HW2_Group<No>.mat`, one per group (see "Results file")
Submission	One per group: report, code, Simulink models (if used) and results file

Late submissions

Late submissions lose 10 points per day and are accepted for up to 3 days after the due date.

Provided material

For the aerodynamics, engine and atmosphere models, you may either call the Simulink models that you built in Homework 1 from your m-files, or write these models as m-file functions. In both cases, use the same data files (f16_AerodynamicData.data, f16_AerodynamicData.BP and f16_EngineData.mat).

For the details of the aerodynamic modeling, see NASA Technical Paper 1538 [1].

Modeling conventions

- **Atmosphere:** use the standard atmosphere model given with the F-16 model in Stevens, Lewis, and Johnson [2], Appendix A.6 (p. 715). This model uses English units; implement it in SI units (altitude in *m*, temperature in *K*, density in *kg/m³*, airspeed in *m/s*, pressure in *Pa*) by converting its constants.
- **Axes:** express the aerodynamic forces (X, Y, Z) and moments (L, M, N) in body axes, with the moments taken about the center of gravity at $x_{cg} = 0.35\bar{c}$.
- **Unit conversions:** the aerodynamic and engine data are in English units. Convert them to SI units using 1 *ft* = 0.3048 *m* and 1 *lbf* = 4.448222 *N*.
- **Units of the results:** give all results in SI units, with angles in *rad* and angular rates in *rad/s*; give the throttle setting δ_T in *%*.
- **Leading-edge flap:** at each trim condition, use the automatic leading-edge flap schedule

$$\delta_{lef} = 1.38\alpha - 9.05\frac{\bar{q}}{p_s} + 1.45$$

where $\bar{q}$ is the dynamic pressure and p_s is the static pressure; α and δ_{lef} are in *deg*. Limit δ_{lef} to the range [0, 25] *deg*.

1. Problems

Problem 1 [50 points] – Trim conditions

Write generic MATLAB code (m-files) that computes the trim conditions for straight flight and for steady-state turning flight. The code must include, or call, the aerodynamics, engine and atmosphere models (see "Provided material"). Define a cost function to be minimized, and minimize it with `fmincon`. Then find the trim conditions in Table 1 at c.g. ($0.25\overline{c}$) for Cases 1–5 and ($0.35\overline{c}$) for Case 6.

Table 1. Trim cases

CASE	Flight definition	Mach	Altitude (*ft*)	γ (*deg*)	β (*deg*)	ϕ (*deg*)	n_z (*g*)
1	General straight flight (level flight)	0.6	10000	0	–	0	–
2	General straight flight (descent)	0.3	3000	-3	–	0	–
3	General straight flight (climb)	0.5	3000	10	–	0	–
4	Steady-state turn	0.6	10000	0	0	60	–
5	Steady-state turn, defined by n_z	0.6	10000	0	0	–	4
6	General straight flight (level flight)	0.6	10000	0	–	0	–

A dash (–) means that the value is not specified for that case.

Required output. Store the trim results of all six cases in `TrimResults` (see "Results file" below).

Problem 2 [50 points] – Numerical linearization

Write generic MATLAB code (m-files) that numerically linearizes the aircraft model around a given trim condition.

Problem 2, part a [25 points] – Linearized model in body axes

Define the linearized model as

$$\boldsymbol{\Delta\dot{x} = A\Delta x + B\Delta u}$$

where

$$\begin{bmatrix} \Delta\dot{u} \\ \Delta\dot{w} \\ \Delta\dot{q} \\ \Delta\dot{\theta} \\ \Delta\dot{v} \\ \Delta\dot{p} \\ \Delta\dot{r} \\ \Delta\dot{\phi} \end{bmatrix} = \begin{bmatrix} \boldsymbol{A11}_{4\times4} & \boldsymbol{A21}_{4\times4} \\ \boldsymbol{A12}_{4\times4} & \boldsymbol{A22}_{4\times4} \end{bmatrix} \begin{bmatrix} \Delta u \\ \Delta w \\ \Delta q \\ \Delta\theta \\ \Delta v \\ \Delta p \\ \Delta r \\ \Delta\phi \end{bmatrix} + \begin{bmatrix} \boldsymbol{B11}_{4\times2} & \boldsymbol{B21}_{4\times2} \\ \boldsymbol{B12}_{4\times2} & \boldsymbol{B22}_{4\times2} \end{bmatrix} \begin{bmatrix} \Delta\delta_T \\ \Delta\delta_h \\ \Delta\delta_a \\ \Delta\delta_r \end{bmatrix}$$

Use body-fixed axes; linearize at c.g. ($0.25\overline{c}$) for (i)–(ii) and at c.g. ($0.35\overline{c}$) for (iii):

i. $M = 0.6, h = 10000\ ft, \gamma = 0°, \phi = 0°$ (steady-state wings-level flight);
ii. $M = 0.6, h = 10000\ ft, \gamma = 0°, \beta = 0°, \phi = 60°$ (steady-state turning flight);
iii. $M = 0.6, h = 10000\ ft, \gamma = 0°, \phi = 0°$ (steady-state wings-level flight).

Required output. Compare the linearized models obtained at the three trim conditions and comment on the differences. Store the matrices in `A_Body` and `B_Body` (see "Results file" below).

Problem 2, part b [25 points] – Linearized model in wind axes

Define the linearized model as

$$\boldsymbol{\Delta\dot{x} = A\Delta x + B\Delta u}$$

where

$$\begin{bmatrix} \Delta\dot{V}_T \\ \Delta\dot{\alpha} \\ \Delta\dot{q} \\ \Delta\dot{\theta} \\ \Delta\dot{\beta} \\ \Delta\dot{p} \\ \Delta\dot{r} \\ \Delta\dot{\phi} \end{bmatrix} = \begin{bmatrix} \boldsymbol{A11}_{4\times4} & \boldsymbol{A21}_{4\times4} \\ \boldsymbol{A12}_{4\times4} & \boldsymbol{A22}_{4\times4} \end{bmatrix} \begin{bmatrix} \Delta V_T \\ \Delta\alpha \\ \Delta q \\ \Delta\theta \\ \Delta\beta \\ \Delta p \\ \Delta r \\ \Delta\phi \end{bmatrix} + \begin{bmatrix} \boldsymbol{B11}_{4\times2} & \boldsymbol{B21}_{4\times2} \\ \boldsymbol{B12}_{4\times2} & \boldsymbol{B22}_{4\times2} \end{bmatrix} \begin{bmatrix} \Delta\delta_T \\ \Delta\delta_h \\ \Delta\delta_a \\ \Delta\delta_r \end{bmatrix}$$

Use wind axes; linearize at c.g. ($0.25\overline{c}$) for (i)–(ii) and at c.g. ($0.35\overline{c}$) for (iii):

i. $M = 0.6, h = 10000\ ft, \gamma = 0°, \phi = 0°$ (steady-state wings-level flight);
ii. $M = 0.6, h = 10000\ ft, \gamma = 0°, \beta = 0°, \phi = 60°$ (steady-state turning flight);
iii. $M = 0.6, h = 10000\ ft, \gamma = 0°, \phi = 0°$ (steady-state wings-level flight).

Required output. Compare the linearized models obtained at the three trim conditions and comment on the differences. Store the matrices in `A_Wind` and `B_Wind` (see "Results file" below).

Results file (one per group)

Each group submits one MAT-file named `HW2_Group<No>.mat`, where <No> is the group number. The file must contain exactly the five variables below, saved as double arrays:

Variable	Size	Content
`TrimResults`	6 × 15	Row k: case k of Table 1. Columns, in this order: $V_T, \alpha, \beta, \phi, \theta, \psi, p, q, r, \delta_T, \delta_h, \delta_a, \delta_r, \delta_{lef}, J$. Column 6 is ψ for Cases 1–3 and 6 and $\dot{\psi}$ for Cases 4–5 (steady-state turns). J is the value of your cost function at the trim condition.
`A_Body`	8 × 8 × 3	Matrix $\boldsymbol{A}$ of Problem 2, part a, with states as defined. Pages 1–3 correspond to conditions i–iii, respectively.
`B_Body`	8 × 4 × 3	Matrix $\boldsymbol{B}$ of Problem 2, part a, with the columns in the order $\Delta\delta_T, \Delta\delta_h, \Delta\delta_a, \Delta\delta_r$. Pages as in `A_Body`.
`A_Wind`	8 × 8 × 3	Matrix $\boldsymbol{A}$ of Problem 2, part b, with states as defined. Pages 1–3 correspond to conditions i–iii, respectively.
`B_Wind`	8 × 4 × 3	Matrix $\boldsymbol{B}$ of Problem 2, part b, with the columns in the order $\Delta\delta_T, \Delta\delta_h, \Delta\delta_a, \Delta\delta_r$. Pages as in `A_Wind`.

All values are in SI units: speeds in *m/s*, angles in *rad*, angular rates in *rad/s*, and the throttle setting δ_T in *%*. Do not change the order of the rows, columns or pages, and do not add other variables, so that the results of all groups can be compared directly. For example:

```
save('HW2_Group<No>.mat', 'TrimResults', 'A_Body', 'B_Body', ...
     'A_Wind', 'B_Wind')
```

2. Reference material

[1] L. T. Nguyen, M. E. Ogburn, W. P. Gilbert, K. S. Kibler, P. W. Brown, and P. L. Deal, *Simulator Study of Stall/Post-Stall Characteristics of a Fighter Airplane With Relaxed Longitudinal Static Stability*, NASA Technical Paper 1538, 1979.

Use for this homework: details of the aerodynamic modeling and the leading-edge flap schedule.

[2] B. L. Stevens, F. L. Lewis, and E. N. Johnson, *Aircraft Control and Simulation: Dynamics, Controls Design, and Autonomous Systems*, 3rd ed., John Wiley & Sons, Hoboken, NJ, 2016.

Use for this homework: trim and linearization, and the standard atmosphere model (Appendix A.6), to be implemented in SI units (see "Modeling conventions").

3. Deliverables

This homework is group work. Each group makes one submission that contains all of the following:

- **Report:** at most 8 pages, submitted through Ninova.
- **MATLAB code:** all the m-files that you have written. The code must run and be easy to follow.
- **Simulink models:** if your code calls Simulink models, submit them saved in MATLAB R2023a.
- **Readme:** a "readme" file for the code, if necessary.
- **Results file:** one MAT-file per group, `HW2_Group<No>.mat`, in the layout given in "Results file".